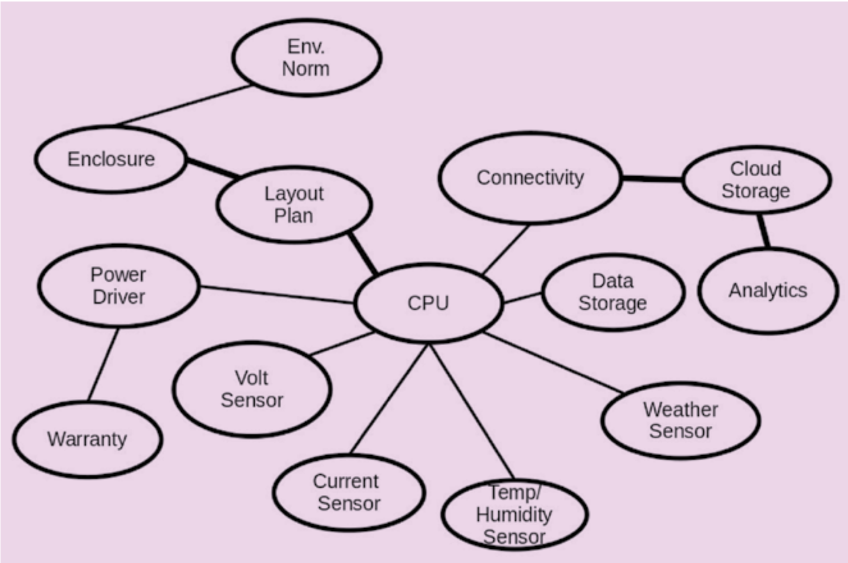

Various factors to be considered for integration

tively. While offering a wide array of options is appealing, the costs and complexity involved could eat into our bottom line.

Striking the right balance between variety and manageability is essential. We want to avoid overwhelming our distribution network and leaving customers unsatisfied due to product unavailability. At the same time, we must consider our financial health in this venture.

Ultimately, the decision we make needs to factor in the convenience of our customers, the efficiency of our operations, and the sustainability of our profits. It is about finding the sweet spot that benefits all stakeholders involved.

Creating product family. Customers' expectations for our products can vary widely. While there is a plethora of product choices available, it is essential that we strategically balance options and expenses.

Modular design plays a crucial role in our flexibility, allowing us to effortlessly integrate or remove features. For instance, consider our cloud-based analytics feature—we can seamlessly offer it alongside the original hardware without any changes. Additionally, we could introduce an add-on weather station, albeit some adjustments to our base design would be necessary.

Speaking of variations, the distinction between our basic model and the simplified temperature and humidity monitoring version lies primarily in the sensors used. The big question is whether we should craft a combined design for both the weather station variant and the basic model or opt for distinct designs for each.

To untangle these decisions, our focus shifts to calculating the overall expenses for each option based on expected sales volume. In this scenario, our market research team projects sales of 2,000 units of the base model, 1,500 units of the model with basic weather monitoring, and 500 units of the model with a separate weather station per month.

Incorporating communication with the weather station requires an RS232-RS485 converter chip, priced at ₹40. The primary difference between the base model and the basic weather station comes from the DHT-11 sensor, valued at ₹70. If we integrate these components into our base circuit, we'll be confined to a single product configuration.

However, our logistical expert emphasises that maintaining separate stocks for each variant would demand about a 15-day supply for each type. Contrarily, a single product type necessitates only 1,000 units in inventory. Our estimated product cost stands at ₹3,500.

In a nutshell, combining all three variants into one assembled product incurs an additional cost of ₹280,000 per month—calculated as ₹40 × (2,000 + 1,500) + ₹70 × 2,000. Opting for separate inventories entails a cost of ₹3,500,000, alongside an additional inventory carrying expense of ₹700,000 due to a 20% capital cost.

The carrying cost of inventory surpasses the component costs. Hence, a pragmatic approach would involve maintaining a singular product configuration while offering the remote weather station and cloud analytics as optional items, possibly on a subscription basis. This approach ensures a harmonious balance between customer choices and efficient resource management.

Previously, we have delved into the art of creating products, from the initial spark of an idea to a thriving product family. These concepts are versatile, and suitable for crafting physical goods, software, or services.

The key to a triumphant product lies in its quality, the affection it garners, and its ability to engage customers consistently, turning into a habit. Products of this calibre and their variations tend to cultivate customer loyalty, standing the test of time. Often, customers are willing to invest a premium for the privilege of owning such cherished products.

In our upcoming and final article, we will explore the last piece of the puzzle: Packaging and the crucial realm of after-sales service. EFY